

# A Survey on Hate Speech Detection using Natural Language Processing

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## Paper Summary

This paper, titled “*A Survey on Hate Speech Detection using Natural Language Processing*,” explores automated methods for identifying hate speech on social media platforms. The exponential rise in hate speech poses challenges for manual moderation, which is time-consuming, inconsistent, and impractical given the sheer volume of data. NLP-based automated detection offers a scalable solution. Hate speech refers to communication that disparages individuals or groups based on attributes such as race, ethnicity, religion, gender, or sexual orientation. Related terms include abusive language and cyberbullying. Definitions vary across cultural and legal contexts, adding complexity to detection. The paper discusses key approaches to hate speech detection, including supervised learning algorithms such as SVMs, Logistic Regression, and Deep Learning, which rely on labeled datasets. Semi-supervised learning methods, including bootstrapping techniques, aim to expand annotated datasets, while multimodal approaches integrate text with images, audio, or video to enhance contextual understanding. Feature engineering techniques include surface features such as N-grams, character-level patterns, and token lengths; word embeddings like Word2Vec and GloVe; sentiment analysis for detecting polarity and intensity; and lexical resources employing dictionaries of slurs and hate-related lexicons.

Datasets used for training and evaluation are often sourced from platforms like Twitter, YouTube, and Instagram. However, challenges persist, including skewed data distributions, inconsistent annotations, and the ambiguity introduced by sarcasm and indirect language. Furthermore, cultural and linguistic diversity complicates detection, as definitions of hate speech vary across regions. Future research focuses on enhancing deep learning models, expanding multilingual datasets, and improving context-aware detection techniques. In conclusion, NLP-based hate speech detection is essential for fostering safer online environments. Continued advancements in model design, dataset diversity, and cultural awareness are critical for addressing existing limitations.